AP PHYSICS C

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0.1 Foreword for anybody else using this

As of writing this, I've already taken AP Physics 1 and 2 and Calc BC, so I'll skim over those stuff. I steal the images from google. If this violates copyright somehow, please email me and I'll remove it.

1 Unit 1: Kinematics

1.1 Scalars and Vectors

Scalars are numbers. Vectors are vectors. Represent as $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}$.

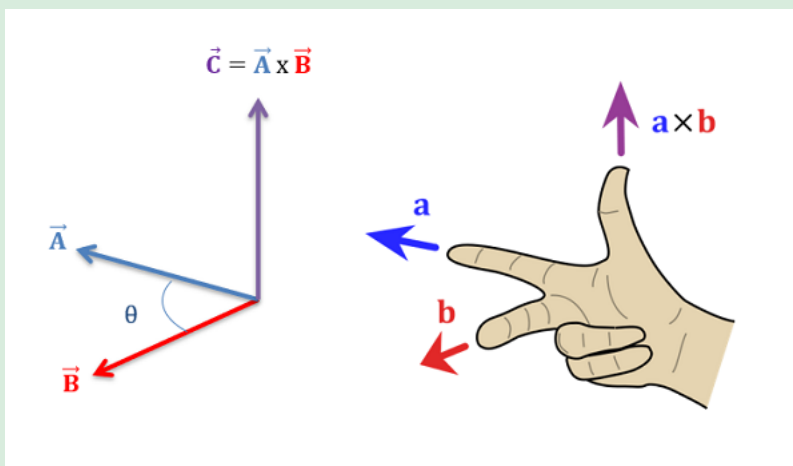
Dot product of two vectors is the length of one of the vectors projected onto the other multiplied by the length of said other vector. Theta = min angle between two vectors.

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$$

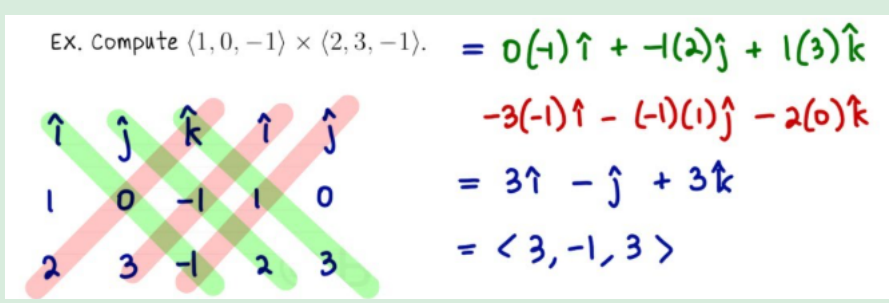
1.1.1 Cross Product

Operation that can be applied to two **three-dimensional** vectors to generate a third vector perpendicular to both vectors. Use the **right hand rule** to deterimine which direction the third vector will go.

Principle 1.1 (Right Hand Rule)
To find the geometrical direction of $\vec{a} \times \vec{b}$, hold your right hand out and point your *index* finger in the direction of $\vec{a}$, and your *middle* in the direction of $\vec{b}$. Your thumb will point in the direction of $\vec{a} \times \vec{b}$.



Principle 1.2 (Calculating the Cross Product)
To calculate the cross product between $\vec{A}$ and $\vec{B}$:



Note the following:

1. $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$ where theta is the minimum angle betwen the two vectors. Also equal to the area of the parallelogram formed by $\vec{a}$ and $\vec{b}$.
2. $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$. This property is known as the **anti-commutativity of the cross product**.
3. $\vec{a} \times \vec{a} = \vec{0}$ $(\theta = 0)$

1.2 Displacement, Velocity, and Acceleration

$$\vec{v} = \frac{\mathrm{d}\vec{s}}{\mathrm{d}t}, a = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\vec{s}}{\mathrm{d}t^2} \implies \Delta\vec{s} = \int \vec{v}(t)dt, \Delta\vec{v} = \int \vec{a}(t)dt$$

Derivatives and intergrals of vector-valued functions can be done component wise.

$$\vec{s} = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k} \implies \frac{\mathrm{d}\vec{s}}{\mathrm{d}t} = x'(t)\hat{\imath} + y'(t)\hat{\jmath} + z'(t)\hat{k}$$

$$\vec{s} = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k} \implies \int_{t_i}^{t_f} \vec{s}dt = \int_{t_i}^{t_f} x(t)dt + \int_{t_i}^{t_f} y(t)dt + \int_{t_i}^{t_f} z(t)dt$$

For constant one-dimensional acceleration, kinematic equations apply:

$$\Delta v = \int adt \implies v_f = v_i + at$$

$$\Delta s = \int v(t)dt = \int (v_i + at)dt \implies \Delta s = v_i t + \frac{1}{2}at^2$$

$$v_f = v_i + at \implies v_f^2 = v_i^2 + 2v_i at + a^2t^2 \implies v_f^2 = v_i^2 + a(2v_i t + at^2) \implies v_f^2 = v_i^2 + 2a\Delta x$$

Falling object experience constant acceleration: $a = g \approx 9.81\mathrm{m\,s^{-2}} \approx 10\mathrm{m\,s^{-2}}$. Projectiles experience constant vertical accleration of gravity, but no horizontal acceleration.

1.3 Representing Motion

Distance, velocity, and acceleration vs. time graphs exist. Yipee!

1.4 Reference Frames and Relative Motion

Let $\vec{v}_{AB}$ = velocity of A relative to B.

$$\vec{v}_{AB} = -\vec{v}_{BA}$$

$$\vec{v}_{AC} = \vec{v}_{AB} + \vec{v}_{BC}$$

These are also known as **Galilean Transformations.** Note these equations will break down near the speed of light due to relativity shit.

Example 1.3
The president's airplane flies at $250\mathrm{m\,s^{-1}}$ to the east with respect to the air. The air is moving at $35\mathrm{m\,s^{-1}}$ to the north with respect to the ground. Find the velocity of the president's airplane with respect to the ground.

Answer 1

Let $\hat{\imath}$ = north and $\hat{\jmath}$ = east.

$$\vec{v}_{PG} = \vec{v}_{PA} + \vec{v}_{AG} = (250\hat{\jmath} + 35\hat{\imath})$$

$$|\vec{v}_{PG}| = \sqrt{250^2 + 35^2} \approx 252.4\,\mathrm{m\,s^{-1}}$$

$$\theta_{PG} = \arctan\frac{35}{250} \approx 7.97^\circ \text{ relative to East}$$

1.5 Motion in Two or Three Dimensions

everything that we've talked about already considers three dimensions :3

2 Unit 2: Force and Translational Dynamics

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8 Unit 8: Electric Charges, Fields, and Gauss's Law

8.1 Electric Charge and the Electric Force

Charge... exists. It's scalar and measured in SI units **Coulombs**.

Coulomb's Law describes the electrostatic force between two charged particles:

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0}\frac{|q_1 q_2|}{r^2} = k\frac{|q_1 q_2|}{r^2}$$

ϵ_0 represents the **electric permittivity of free space** and has a constant value (units $C^2N^{-1}m^{-2}$). Electric permittivity measures "the degree to which a material or medium is polarized in the presence of an electric field" and is determined by "the ease with which electrons can change configurations within a given material." Electric polarization can be modeled as the induced rearangement of electrons by an external electric field.

8.2 Conservation of Electric Charge and the Process of Charging

Like charges repel. Charges on system influence each other.

8.3 Electric Fields

Electric fields exist. $\vec{E} = \frac{\vec{F_e}}{q_e} = k\frac{q}{r^2}\hat{r}$

Note while in electrostatic equilibrium (no net movement of charges), the excess charge of a conductor is distributed on the surface of said conductor, and the electric field within said conductor is 0. Note that this is **not** true for insulators: charge is distributed through the surface AND interior, and the electric field within may be non-zero.

8.4 Electric Fields of Charge Distributions

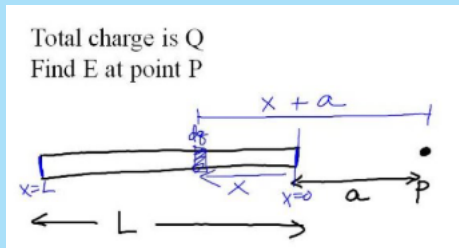
For continious charge distributions, you can consider tiny point charges dq and apply Coulomb's Law to each of them, then add their effects together. The fact that you can add them together is known as **the principle of superposition.** Mathematically:

$$\vec{E} = k\int \frac{\mathrm{d}q}{r^2}\hat{r}$$

You can typically directly solve for Q or express the integral in terms of dr using **linear charge density** $\lambda = \frac{\mathrm{d}Q}{\mathrm{d}L} \approx \frac{Q}{L}$, **surface charge density** $\sigma = \frac{Q}{A}$, or **volumetric charge density** $\rho = \frac{Q}{V}$

Example 8.1

Determine the electric field at point P.



Answer 2

$$\lambda \, \mathrm{d}r = \mathrm{d}Q$$

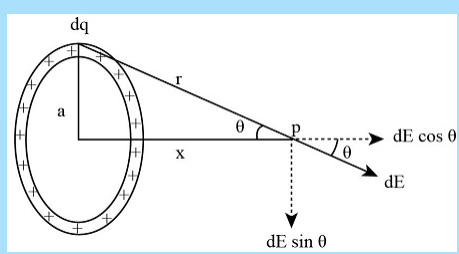
$$\begin{aligned}
\vec{E} &= k \int \frac{\mathrm{d}Q}{r^2} \hat{r} \\
&= k\hat{r} \int_a^{a+L} \frac{\lambda \, \mathrm{d}r}{r^2} \\
&= \lambda k \hat{r} \left(\frac{-1}{a+L} + \frac{1}{a} \right) \\
&= \lambda k (\hat{\imath}) \left(\frac{L}{a(a+L)} \right) \\
&= \boxed{\frac{Qk\hat{\imath}}{a(a+L)}}
\end{aligned}$$

Note when $a >> L$, it degenerates the the below, which suggests that it acts as a point charge at sufficently large distances.

$$\vec{E} \approx \frac{Qk\hat{\imath}}{a(a^2)}$$

Example 8.2

Determine the electric field at P.



Answer 3
Note that "horizontal" $\mathrm{d}\vec{E}\sin\theta$ components cancel via symmetry. Thus, you only need to consider components proportional to $\cos\theta$.

$$\begin{aligned}\vec{E} &= k\int \frac{\mathrm{d}Q}{a^2+x^2}\hat{\imath}\cos\theta \\ &= k\hat{\imath}\int \frac{\mathrm{d}Q}{a^2+x^2} \times \frac{x}{\sqrt{a^2+x^2}} \\ &= \boxed{\frac{k\hat{\imath}Qx}{(a^2+x^2)^{1.5}}}\end{aligned}$$

8.5 Electric Flux

Flux describes the amount of a given quantity that passes through a given area. **Electric flux** Φ_E describes the amount of electric field that passes through a defined area. Note the area vector is defined to be outward from a closed surface.

$$\Phi_E = \vec{E}\cdot\vec{A}$$

For an arbitrary surface:

$$\Phi_E = \int \vec{E}\cdot\mathrm{d}\vec{A}$$

8.6 Gauss's Law

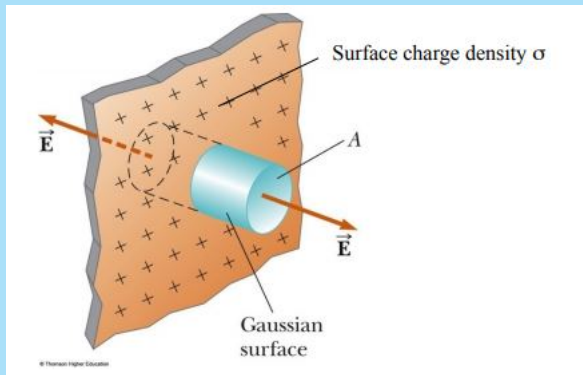
Definition 8.3 (Gauss's Law)
For an arbitrary **Gaussian Surface** (3d closed surface):

$$\Phi_E = \oint \vec{E}\cdot\mathrm{d}\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

Usually you can rig the surface such that the electric field is perpendicular or parallel to all components of the area, with a sphere or a cylinder.

Example 8.4

Determine the electric field surrounding an infinitely large, thin plane of positive changes with uniform surface charge density.



Answer 4

Set up the Gaussian surface as shown. The area vector is perpendicular to the electric field along the "wrapping" of the cylinder. The area vector is parallel to the electric field in the same direction for the two "caps" of the cylinder. Thus,

$$2EA = \frac{q_{enc}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0} \implies E = \boxed{\frac{\sigma}{2\epsilon_0}}$$

Note for the case of a two "infinite" plane capacitor, we assume the field outside the capacitor cancels. Thus,

$$E_{capacitor} = \frac{\sigma}{\epsilon_0}$$

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